

QB4AIRA

A Question Bank for Responsible AI Risk Assessment

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// We introduce QB4AIRA, a question bank developed by refining questions from five globally recognized AI risk frameworks, categorized based on Australia's AI ethics principles. QB4AIRA is a valuable resource for stakeholders in assessing and managing AI risks, paving the way for new risk frameworks and guidelines. //



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ARTIFICIAL INTELLIGENCE (AI) has witnessed remarkable advancements in recent years, permeating numerous domains and transforming

various aspects of our lives. However, this rapid advancement of AI technologies has also raised concerns about their responsible development

and application. For instance, While generative AI models (e.g., large language models like GPT-4) have demonstrated impressive capabilities in generating text, images, audio, and code, they also present significant limitations and risks, such as hallucination, generating harmful content, and users' overreliance.¹ Incidents documented in repositories (e.g., the AI Incident Database^a recording more than 3,500 AI incidents as of August 2024) highlight the urgency of addressing AI risks.

To ensure the responsible development, deployment, and utilization of AI systems, the concept of responsible AI (RAI) has emerged. RAI involves the practice of developing and utilizing AI systems that benefit individuals, groups, and society as a whole, while minimizing the risk of adverse consequences.^{2,3} Risks, in this context, refer to the likelihood and consequences of AI systems failing to align with RAI principles, like fairness, privacy, explainability, and accountability. Ensuring adherence to these principles throughout AI development is therefore crucial.

Several organizations have developed principle-driven RAI risk assessment frameworks to guide the implementation of RAI principles, such as the U.S. National Institute of Standards and Technology (NIST) AI Risk Management Framework^b and the European Union Assessment List for Trust AI Framework.^c While these frameworks provide valuable guidance for evaluating risks of AI systems, there is a lack of a comprehensive and well-structured AI risk assessment question bank that incorporates interconnected and

^a<https://incidentdatabase.ai/>

^b<https://www.nist.gov/itl/ai-risk-management-framework>

^c<https://futurium.ec.europa.eu/en/european-ai-alliance/pages/welcome-altai-portal>

layered questions and connectivity to other standards and frameworks.³ While questions are only a part of the whole effort toward RAI, having relevant and well-structured questions

identification of 143 academic works and 222 pieces of grey literature. Through iterative reviews involving two additional researchers, this pool was refined to 16 frameworks, which

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is crucial for effectively assessing and mitigating risks associated with AI systems. However, existing question banks often focus on specific aspects, overlooking the holistic nature of RAI principles. For instance, Liao et al.^{4,5} introduced a question bank specifically focused on transparency and explainability, neglecting other crucial principles. Additionally, the U.S. Department of Energy's AI Risk Management Playbook^d provides extensive guidance on identifying AI risks and recommending mitigations, but it lacks integration with diverse standards, frameworks, and related resources.

To bridge this gap, we propose QB4AIRA as a comprehensive and connected resource for AI risk assessment. In this article, we present the development of QB4AIRA, which is built on existing frameworks identified in a previous multivocal literature review conducted by our research team, as detailed in Xia et al.³ The initial review was led by one researcher, with support and guidance from two senior researchers, resulting in the

were then analyzed for their core characteristics and alignment with RAI principles. We selected five frameworks among these frameworks based on the review results, which is based on 1) global recognition, 2) inclusion of risk assessment questions, and 3) representation of different regions and industry leaders. These frameworks encompass various geopolitical contexts, including the E.U. Trustworthy AI Assessment List, Canada Algorithmic Impact Assessment,^e Australia NSW AI Assurance Framework,^f Microsoft Responsible AI Impact Assessment Guide,^g and the U.S. NIST AI Risk Management Framework.

The development of QB4AIRA was a systematic and iterative process. Initially, two researchers collected 382 risk questions from the five frameworks. Four researchers then reviewed and validated the

results. Through iterative refinement, we condensed 382 questions into 293. After this refinement, we took an additional step to analyze and structure these questions. We systematically categorized the questions based on Australia's AI ethics principles,⁶ and conducted concept mapping to identify common risk themes.⁷ Each question was mapped to relevant development phases and organized into three hierarchical levels. The hierarchy of risk questions was structured using a decision tree,⁸ considering their level and sequence. This postmortem analysis and categorization into themes and sub-themes were crucial for enhancing the usability and effectiveness of the question bank. QB4AIRA has undergone evaluation through a case study and improved based on the results.

The Question Bank: QB4AIRA

QB4AIRA is a collection of risk questions aimed at supporting the ethical development and use of AI systems. It is designed to meet the requirements of diverse users, including individuals at the C-level, project manager level, developer level, and beyond.

While the question bank includes nearly 300 risk questions across 31 risk themes, its comprehensiveness is demonstrated not only by its coverage of all eight AI ethics principles, but also by its systematic integration of lower-level, detailed risk questions with higher-level ones and related themes. This interconnected approach ensures that the assessment process is not siloed or fragmented, allowing for a cohesive evaluation across different levels of the organization. Unlike existing AI frameworks that often focus on a subset of key principles, QB4AIRA uniquely addresses the full spectrum of ethical considerations.

^d<https://www.energy.gov/ai/doe-ai-risk-management-playbook-airmp>

^e<https://www.canada.ca/en/government/system/digital-government/digital-government-innovations/responsible-use-ai/algorithmic-impact-assessment.html>

^f<https://www.digital.nsw.gov.au/policy/artificial-intelligence/nsw-artificial-intelligence-assurance-framework>

^g<https://blogs.microsoft.com/wp-content/uploads/prod/sites/5/2022/06/Microsoft-RAI-Impact-Assessment-Guide.pdf>

For example, while the International Organization for Standardization/ International Electrotechnical Committee (ISO/IEC) AI management standard^h emphasizes accountability, and the E.U. AI Actⁱ focuses on legal requirements related to accountability, transparency, and privacy, QB4AIRA integrates all eight principles in a way that we haven't seen in other frameworks. This integration enhances the depth and connectivity of the assessments, making QB4AIRA a more holistic tool for evaluating AI risks.

QB4AIRA also tackles the distinct challenges and considerations linked to AI risk assessment, such as unforeseen consequences and the opacity of AI models. It addresses privacy and bias issues stemming from the nature of AI training data (e.g., sensitive data and big data), and helps identify areas where risk management practices may be lacking.

Architecture

- *AI ethics principle*: As shown in Figure 1, we have chosen Australia's ethical AI principles^j as they closely align with similar principles^{9,10} from around the world. These principles serve as the overarching categories for the question bank covering holistic aspects of RAI, such as human, societal, and environmental well-being; human-centered value; fairness; privacy and security; reliability and safety; transparency and explainability; contestability; and accountability.
- *Categories*: Each principle is further divided into multiple categories representing a specific aspect of the principle. The categories act as common risk themes that are clear, specific, and relevant to the AI ethics principle they

represent. Each category contains one or more subcategories, providing even more granularity and specificity to the questions and facilitating a better understanding of the different dimensions of each category.

- *Risk question*: There is a set of questions associated with each category (examples in Table 1). These questions are to identify potential risks of AI systems, ensuring comprehensive scrutiny and resolution of all potential risks.
- *Question ID*: This is to uniquely identify each question within the question bank. Internal IDs are employed for managing the questions internally, including tracking the source framework of each question. All original questions (a total of 382) have been assigned internal IDs. Additionally, the refined questions (293 in total) have been assigned global IDs, which is a universal reference number for external entities.

^h<https://www.iso.org/standard/81230.html>

ⁱhttps://www.europarl.europa.eu/doceo/document/TA-9-2024-0138-FNL-COR01_EN.pdf

^j<https://www.industry.gov.au/publications/australias-artificial-intelligence-ethics-framework/australias-ai-ethics-principles>

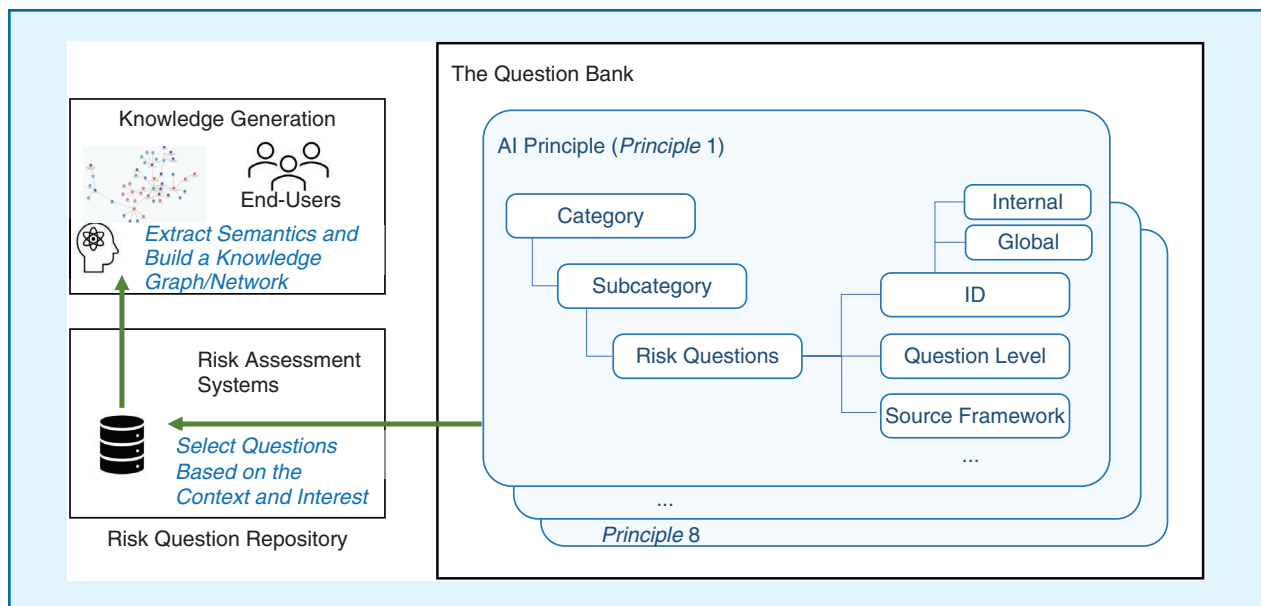


FIGURE 1. The architecture of the question bank.

Table 1. The number of common risk themes (category and subcategory) and questions categorized by AI ethics principles.

Principle	#Category	#Subcategory	#Question	#Framework
Human well-being	5	12	71	5
Human values	3	5	19	4
Fairness	3	5	38	5
Privacy/security	4	7	46	5
Reliability/safety	5	15	49	4
Transparency/explainability	4	7	30	5
Contestability	2	4	4	2
Accountability	5	13	36	5
	31	68	293	2–5
(Examples of risk themes and questions) L1: level-1 question, L2: Level-2 question				
Accountability	Auditability	Traceability/ logging	Do you establish mechanisms that facilitate the system's auditability, such as ensuring traceability and logging of the AI system's processes and outcomes? (L1)	European Union
			Does the audit trail identify the authority or delegated authority identified in legislation? (L2)	Canada
		Independent audit	Do you ensure that the AI system can be audited independently? (L2)	European Union
	Redress	Redress mechanism	Do you establish an adequate set of mechanisms that allows for redress in case of the occurrence of any harm or adverse impact? (L1)	European Union
			Do you put mechanisms in place both to provide information to (end-)users/ third parties about opportunities for redress? (L2)	European Union
Transparency/ explainability	Traceability	Traceability mechanism	Do you establish measures that can ensure traceability? (L1)	European Union
			What is the extent of the risk associated with incomplete documentation of AI system design, or implementation, or operation? (L2)	New South Wales
	Explainability	System explainability	Can the AI system produce outcomes that all users can understand? (L1)	European Union
			Do you explain, validate and document the AI model and the output is interpreted within its context to inform responsible use and governance? (L2)	NIST
		Assess explainability	Do you assess to what extent the decisions and hence the outcome made by the AI system can be understood? (L2)	European Union
Reliability/safety	Resilience	Mechanism	Do you have organizational policies and practices to foster a critical thinking and safety-first mindset in the design, development, deployment, and uses of AI systems to minimize potential negative impacts? (L1)	European Union
	Accuracy	Data quality	Do you put in place measures to ensure that the data used is comprehensive and up to date? (L2)	European Union
			Do you put measures in place to assess whether there is a need for additional data, for example to improve accuracy or to eliminate bias? (L2)	European Union
		Limitations	Are there conditions in the deployment environment that would affect the system's performance? (L3)	Microsoft

There are also selected examples of the risk themes and questions for accountability, transparency and explainability, and reliability and safety.

- *Question level:* Each question has a question level (1, 2, or 3), indicating its importance and level of detail within the risk category. The question level is essential for facilitating tiered risk assessment (discussed further in the “Case Study: AI System Risk Assessment” section).
- *Source framework:* This provides a link to the original framework of each question, which allows users to easily access and verify the original source of the questions. We indicate the extent to which each subcategory is addressed as a common concept across frameworks by including the number of source frameworks, ranging from one to five.
- *External entities:* QB4AIRA can be utilized by risk assessment systems or individual users. They can access QB4AIRA via global IDs and select the most appropriate ones based on their specific use case, context, and interests.

Key Features

Concept Map. We employed a concept mapping approach⁷ to analyze and group related risk questions, thereby identifying AI risk themes. This process clarifies the relationships between different risk themes and provides a comprehensive overview of the key areas of concern for AI systems.

Table 1 provides an overview of the identified risk themes. With a total of 293 questions, QB4AIRA covers 31 risk themes and 68 subthemes, demonstrating its comprehensive coverage of various aspects of RAI.

In **Figure 2**, the concept map represents the hierarchical organization of the risk themes and their distribution across eight AI principles. Each principle consists of two to five

concepts that represent crucial areas of risk requiring responsible assessment and management in AI. This indicates that these risk areas should be ensured and controlled throughout the entire life-cycle of AI systems.

feasible to ask all of them. The level of questions has been determined based on their relative importance and specificity within the category. This accommodates the needs of different groups of stakeholders

While questions are only a part of the whole effort toward RAI, having relevant and well-structured questions is crucial for effectively assessing and mitigating risks associated with AI systems.

The expanded concept map (the outermost circle in **Figure 2**) delves into 68 subthemes within each primary risk category for in-depth exploration and understanding of specific risks. Highlighted themes indicate subthemes that are derived from three or more source frameworks. The superscript number next to each theme represents the number of questions associated with it. These highlighted themes suggest common coverage across multiple frameworks and highlight areas that warrant more attention in the risk assessment process. Similarly, themes with a higher number of associated questions indicate greater potential risks, underscoring their importance in risk management. This approach supports evidence-based decision-making in prioritizing risk areas.

Tiered Risk Assessment. The notion of tiered risk assessment arose from the interest in examining risks at different levels of detail while effectively utilizing the large number of questions available, as it may not be

in the risk assessment process and provides a top-down approach that allows organizations to focus on the most important and high-level risks first.

The top-level question in each category is regarded as the most important and fundamental question that provides a broad overview of the key risks. These questions are primarily intended for high-level decision-makers, such as C-levels.

Subsequent questions are organized into second and third levels. Second-level questions are typically more specific than the top-level question and help to provide additional detail and context. They target senior managers and individuals with specific roles in overseeing AI-related projects. Third-level questions are even more detailed and represent subquestions that help to further break down the key risks. These questions are aimed at practitioners who require a detailed understanding of the technical and operational aspects of AI risk assessment.

For example, one of the top-level questions for transparency and explainability, “Does the outcome result in something that all users can understand?”, is an import and high-level question. However, it does not delve into specific considerations and risks in the system’s development. A possible second-level question is, “Is the AI system designed to be inherently explainable from the outset?” This question assesses whether the project team considers

“transparency by design” and manages potential design issues in the early development stages. As such, this question is suitable for the management-level stakeholders rather than C-levels. “Do you research and try to use the simplest and most interpretable model possible for the AI system?” can be used to further explore the topic if the development team has conducted sufficient research on interpretability during the model selection process.

Streamlined Risk Assessment. It is essential to streamline the risk assessment process and enhance efficiency while maintaining its effectiveness in identifying and addressing potential risks.¹¹ We have developed a hierarchical structure using a decision tree, which effectively and efficiently selects the most relevant risk questions from the extensive pool.⁸

Figure 3 depicts the process of exploring risk questions (steps 1–3) based on a preorder traversal approach.¹² This

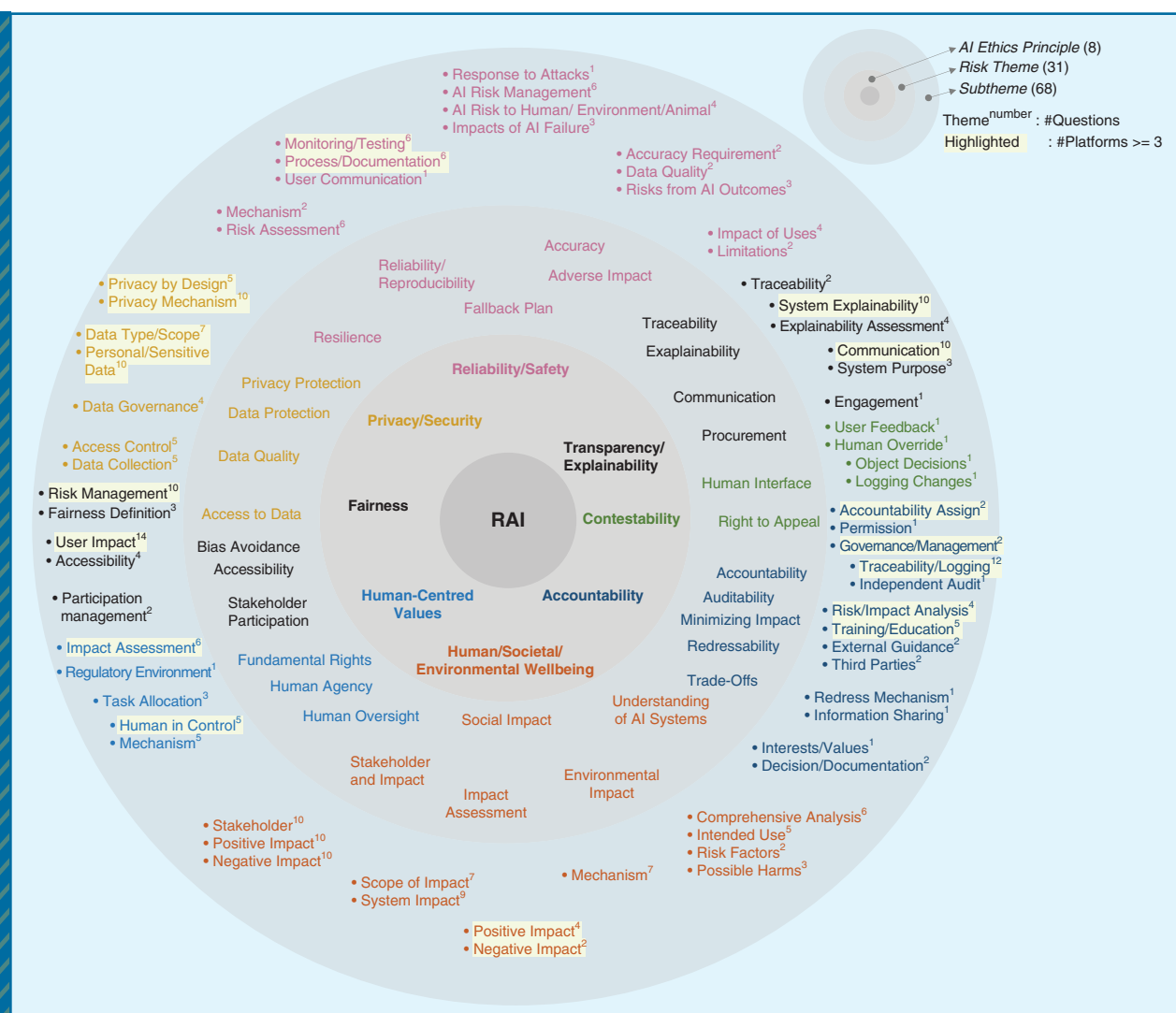


FIGURE 2. Risk concept map for RAI.

initiates by visiting the root node, enabling the determination of which principles should be assessed. It then moves to the first left subtree before proceeding to the right subtrees. Within each subtree, based on the scope of the risk category (the first decision node of the subtree being visited), the top-level question takes precedence (level 1). The leaf nodes of the decision tree indicate whether the risk assessment should be performed solely on the selected categories or on all categories encompassed in the tree, depending on the decision made at the preceding nodes.

Case Study: AI System Risk Assessment

Overview

We have assessed AI ethical risks for eight AI projects selected within a national research organization: PR1 to PR8. These projects primarily focus on research and include AI as a

component within a larger research framework. The projects have different objectives; for example, PR1 aims to provide a comprehensive design tool for AI and PR3 uses AI for drug discovery. Since the projects are currently in their early stages, they present an ideal opportunity to incorporate best practices in RAI for future development.

The project teams had a relatively flat hierarchy with a project lead managing the entire research project, multiple work package leads overseeing sections of the project, and researchers reporting to the work package leads. This case study aimed to examine 1) effectiveness and 2) the feasibility of our question bank in assessing AI risks.

Risk Assessment

The first round of risk assessment sessions occurred between January and February 2023, with each 1.5-h session (10-min for introduction and 10-min/principle) involving

a risk assessor and one to two observers (the authors and external experts) and one to four interviewees, including project leads and, in most cases, work package leads. Initial interviews focused on high-level (levels 1 and 2) questions from QB4AIRA (three to five questions) for each AI principle. These interviews were recorded, transcribed, and rechecked for accuracy. The second round of interviews was conducted a few months later to delve deeper into the identified risks using level 3 questions. The collective interview data were analyzed to assess AI risks in the eight projects.

Results

Effectiveness of the Question Bank.

The question bank's structured approach, combining high-level principles with the different-level questions, has proven to be effective in uncovering potential risks across diverse

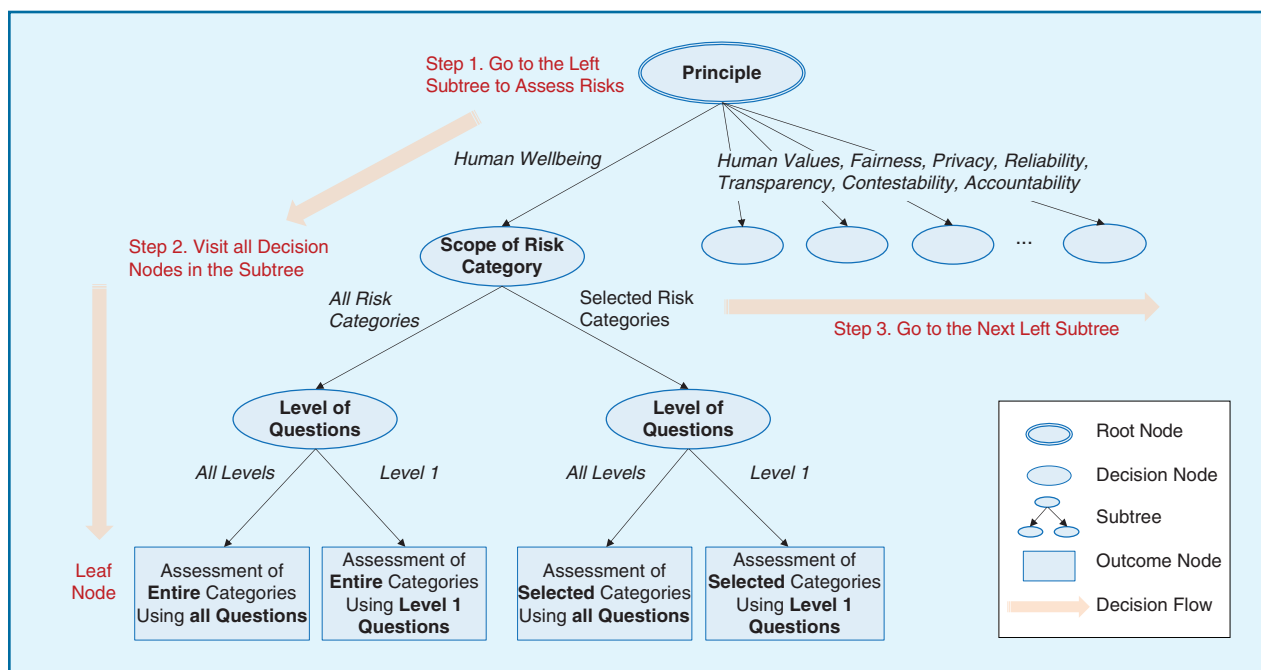


FIGURE 3. Overview of the decision tree for AI risk assessment.

AI projects. We have ensured that the question bank addresses a broad range of risk factors and provides actionable insights. For example, it prompted interviewees to consider overlooked aspects, such as intellectual property risks in AI for drug discovery (PR3). Through the first round of interviews, we identified that accountability posed the highest combined medium to high risk due to a lack of responsible documentation and traceability. Transparency/explainability also presented significant risks due to insufficient clarity in explaining AI outcomes. Conversely, human-centered values, human, societal and environmental well-being, and contestability were found to be relatively low-risk, showing better alignment with ethical practices. The second round of interviews confirmed these findings with the project teams, validating that the question bank effectively highlights critical risk areas and provides a solid foundation for risk management.

Feasibility of the Question Bank. We created a risk register template, including risk ID, category, description, causes, and mitigation measures, based on the question bank. This template adhered to the identified risk themes and utilized three question levels from the question bank to formulate interview questions. We analyzed the projects to understand their contexts and selected three to five questions for each AI ethics principle to prepare interviews. We implemented AI risk assessments, incorporating the risk template, a well-known 3×3 risk assessment matrix, and calculations for impact and probability for risk levels. Through the risk assessment with eight projects, we ensured that the question bank serves as a foundational resource for AI risk assessment, seamlessly integrating into traditional

risk assessment workflows without requiring extensive modifications.

However, several challenges were encountered. First, in early-stage projects with limited model accuracy insights and minimal real-world user engagement, assessing risk levels was challenging. Some risk definitions and questions may not be directly applicable to these projects until they reach commercialization. Second, the effectiveness of the interviews depended heavily on the interviewees. The flat hierarchy of the project team ensured comprehensive responses, which may differ in commercial settings. Last, time constraints necessitated iterative assessments, focusing on more relevant principles. Prioritizing ethical principles based on preliminary information could improve efficiency. Despite these challenges, our question bank has proven to be a feasible tool for AI risk assessment, effectively integrating into existing workflows while highlighting areas for future improvement and refinement.

Feedback and Improvement

Following the risk assessment process, stakeholders provided valuable feedback. They acknowledged that while the QB4AIRA questions were useful in identifying risks, some were not directly relevant to the current project stages. Consequently, stakeholders recommended improving QB4AIRA's structure by incorporating a development stage dimension. Based on this feedback, we reviewed and refined the question bank, mapping questions to different development phases. This enhancement enables a more efficient and effective risk identification process, aligning questions with each project's specific stage. By tailoring questions to the development stage, potential risks

can be identified more accurately and in a manner more applicable to the project's context. This not only enhances the relevance and usefulness of QB4AIRA, but also optimizes resource allocation and efforts in assessing AI risks. This refinement addresses the challenge of assessing risk levels in early-stage projects with limited model accuracy insights and minimal real-world user engagement. By aligning questions with specific development stages, we ensure that the questions are more applicable and actionable throughout the project lifecycle. Additionally, this approach mitigates the impact of time constraints by prioritizing the most relevant ethical principles based on the project's current phase, ultimately leading to a more efficient and focused risk assessment process.

QB4AIRA was developed via collecting and refining questions from five extant AI risk frameworks; it includes 293 questions that cover a wide range of AI risk areas and can be used by different stakeholders, including AI developers, system operators, policy-makers, and regulators. To prioritize and align the questions in QB4AIRA with the context of AI systems, we assessed and determined their level of importance and details. This process aimed to ensure efficient and effective risk assessment while addressing specific needs. By providing a standardized approach to AI risk assessment, QB4AIRA can help ensure that AI systems are developed, deployed, and used in a responsible and ethical manner that addresses potential risks and harms. Furthermore, it can serve as a valuable resource for the development of new AI risk frameworks and guidelines in the future.



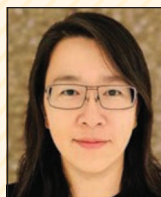
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This study has some limitations and proposes areas for future improvement. The evaluation of the question bank heavily relied on a single case study and participants'

feedback, which may not capture the full spectrum of AI applications and contexts. To address this limitation, future work should include evaluations across a more diverse

range of projects and industries. Additionally, it would be beneficial to compare the proposed framework with existing AI risk assessments to highlight the holistic nature of our



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solution. By evaluating a given project with other existing frameworks, we can identify the differences in risks captured and further validate the effectiveness of our approach. In addition, our future plans include incorporating additional AI risk frameworks, regulations, and standards, such as the E.U. AI Act^k and ISO for AI risk^l to further enhance the coverage and relevance of the question bank. 📄

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^l<https://www.iso.org/standard/77304.html>